Algebra
Class Work ANSWER KEY
Simple Interest vs Compound Interest

Name ______________________________

Date ______________________________

Simple Interest vs Compound Interest

Simple Interest can be calculated by (*principal*) × (*rate*) × (*time*) Formula: $I = prt$

Example:

Find the interest earned by investing \$2,400 at 3% interest rate for 5 years.

$I = prt$
$I = (2400)\ (0.03)\ (5)$
$I = \$360$

Find the future value of the investment after 5 years.

Future Value = principal + interest
Future Value = 2400 + 360
Future Value = \$2730

Compound Interest is calculated by the same formula, except that the interest is calculated at the end of each interest period and then the interest is added to the principal. Compound interest is calculated at the end of each subsequent investment period and the interest continues to be added to the principal.

Exercise 1

You put \$3000 in the bank at 2% APR that is compounded annually. How much money do you have in the account after 6 years? Finish the table to complete this problem.

Year	Beginning Balance	Interest Calculation	Interest	Ending Balance
1	3000	(3000)(.02)	60.00	3060.00
2	3060.00	(3060.00)(.02)	61.20	3121.20
3	3121.20	(3121.20)(.02)	62.42	3183.62
4	3183.62	(3183.62)(.02)	63.67	3247.29
5	3247.29	(3247.69)(.02)	64.95	3312.24
6	3312.24	(3312.24)(.02)	66.24	3378.48

What would be the future value of the investment if *simple interest* was paid?

$I = (3000)(0.02)(6)$
$I = 360$
$FV = 3000 + 360$
$FV = \$3360$

Exercise 2

You put $4000 in the bank at 3% APR that is compounded annually. How much money do you have in the account after 6 years?

NOTE: We'll shorten the process by finding 103% of the principal at the end of each investment period.

Month	Beginning Balance	Ending Balance Calculation	Ending Balance
1	4000	(4000)(1.03)	4120.00
2	4120.00	(4120) (1.03)	4243.60
3	4243.60	(4243.60) (1.03)	4370.91
4	4370.91	(4370.91) (1.03)	4502.04
5	4502.04	(4502.04) (1.03)	4637.10
6	4637.10	(4637.101) (1.03)	4776.21

Exercise 3

1. You inherit $120, 000 from your rich Uncle Bucks. If you invest the money in a CD that pays 4% interest, how much money will you have in four years?

   a. Compute the problem using simple interest. Show work. $I = prt$

$$I = (120000)(0.04)(4)$$
$$I = 19200$$
$$FV = 120000 + 19200$$
$$FV = \$139,200$$

   b. Compute the problem if the interest is compounded annually.

Year 1: (120,000) (1.04) = 124,800
Year 2: (124,800) (1.04) = 129,792
Year 3: (129,792) (1.04) = 134,983.68
Year 4: (134,983.68) (1.04) = 140,383.0272

ANS: $140,383.03

2. Say you put \$10 in a savings account and the account pays 5% *simple interest* for each year that the money is in your account.
   a. Write a formula that could be used to determine the value of your account after x number of years.

      $f(x) = (10)(0.05)x + 10$ or $f(x) = 0.5x + 10$

   b. Describe the graph of the function.

      ANS: The graph is linear. Its slope is 0.05 and its y-intercept is 10.

   c. Use the formula to find the value of the account after 24 years.

      $f(24) = (10)(0.05)(24) + 10$ or $f(x) = (0.5)(24) + 10$
      $f(x) = 22$ $f(x) = 22$

3. Say you put \$10 in a savings account and the account pays 5% interest for each year the money is in the account but the interest is *compounded annually*.

   a. What would be the value of the account after 4 years?
      10 (1.05) = 10.50
      10.50 (1.05) = 11.025
      11.025 (1.05)= 11.57625
      11.57625 (1.05) = 12.1550625 ANS: \$12.16

   b. Describe the graph of the function.

      ANS: The graph is an exponential curve.

   c. Write a formula that could be used to determine the value of the account after x number of years.
      $f(x) = 10(1.05)^x$

   d. Try the formula to find the value of the account in 4 years. Then, try the formula to find the value for 24 years.
      $f(4) = 10(1.05)^4$ $f(24) = 10(1.05)^{24}$
      f(4) = 12.1550625 f(24) = 32.25099944
      ANS: \$12.16 ANS: \$32.25

Future Value for Compound Interest

$$FV = P\,(1 + R)^T$$

Exercise 4

Show work for each problem using the future value formula. Give your final answers to the *nearest cent*.

1. You win a $1000 in a radio contest. If you put the $1000 into an account that pays 3.5% APR *compounded annually*, what will be the value of your investment after 20 years?

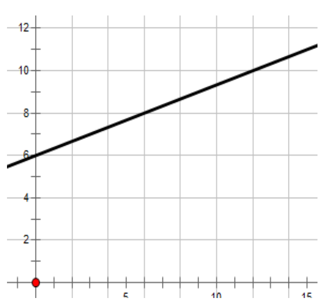
   FV = 1000 (1.035)20
   FV = 1989.788863
   ANS: $ 1989.79

2. Remember the $120, 000 that you inherited from your rich Uncle Bucks? If you invest the money in a CD that pays 4% interest *compounded annually*, how much money will you have in forty years?

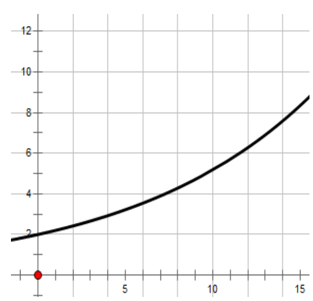
   FV = 120000(1.04)40
   FV = 576122.4754
   ANS (40 years): $576,122.48

Exercise 5

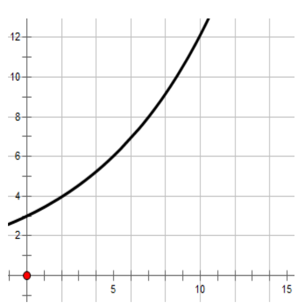
Label each graph to identify whether the graph could represent a *simple interest* problem or a *compound interest* problem.



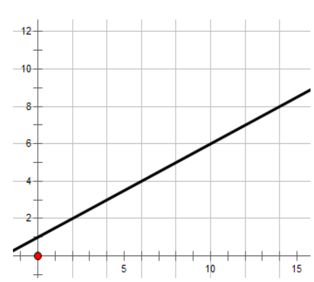
Simple Interest



Compound Interest



Compound Interest



Simple Interest